Practice — The Three States of Matter

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 · MCQ **[1]**

Which state has a definite volume but NO definite shape?

(a) solid (b) liquid (c) gas (d) none

Q2 · MCQ **[1]**

Interparticle forces are **strongest** in —

(a) gases (b) liquids (c) solids (d) all are equal

Q3 · MCQ **[1]**

The temperature at which a solid melts to a liquid at atmospheric pressure is its —

(a) boiling point (b) melting point (c) freezing range (d) dew point

Q4 · MCQ **[1]**

Liquids and gases are together called —

(a) solids (b) fluids (c) crystals (d) vapours

Q5 · ASSERTION–REASON **[1]**

A: Gases have neither a fixed shape nor a fixed volume.
R: The interparticle attractions in gases are negligible, so particles move freely in all directions.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK [1]

The slow change of a liquid to vapour at the surface, below boiling, is called _________.

____

Q7 · FILL IN THE BLANK [1]

In solids, particles can only _________ about fixed positions.

____

Short answer (2 marks each)

Q8 [2]

Give two properties each of solids, liquids and gases (shape and volume).

Q9 [2]

Define melting point and boiling point.

Q10 [2]

Differentiate between boiling and evaporation.

Long / application (3 marks each)

Q11 **[3]**

Explain, in terms of particles and interparticle forces, what happens when a solid is heated until it becomes a liquid and then a gas.

Q12 **[3]**

Why does the same water poured into containers of different shapes keep the level at 200 mL but change shape? What does this tell us about liquids?

Check yourself → Answer key